

The role of political ideology in media coverage of science *

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Abstract

We present four facts about the role of political ideology in media coverage of 125,284 scientists using a novel measure of ideology. First, media outlets are more likely to cover scientific articles written by ideologically aligned scientists. Second, non-ideological quality is strongly associated with coverage decisions—substantially more so than ideological alignment. Third, the association between coverage and scientist ideology is stronger in fields of science where research engages with contemporary political debates, such as history and economics. Fourth, the political relevance of a field can change over time, as we demonstrate using a case study of research on the COVID-19 pandemic.

JEL Codes: O33, L82, P00, I23

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1 Introduction

As scientific research becomes increasingly specialized and technical, audiences rely on media outlets to learn about new research findings [Pew Research Center, 2017]. Effective media coverage of science allows key end users—politicians, business leaders, and technologists—to easily access valuable scientific information [Yin et al., 2022]. It is well-known that ideological slant influences media coverage decisions outside of science (see review in Puglisi and Snyder Jr 2015). Prominent commentators and politicians have argued that the media also slants their coverage of scientific research.¹ Such ideological distortions, if present, erode the social value of science by reducing the quality of information available for decision makers.

In this paper, we present new empirical facts on political ideology and media coverage of scientists. To do this, we develop a novel large-scale measure of scientist ideology based on political donations. We use our measure to investigate whether media outlets disproportionately cover papers written by ideologically aligned researchers. We then ask whether this “ideological preference” leads outlets to cover low-quality science. Finally, we study the extent to which the role of ideology depends on the political salience of research topics by exploring heterogeneity across fields of science and over time. We present a case study of media coverage of the COVID-19 pandemic as a concrete illustration.

To answer our research questions, we developed a novel measure of scientists’ left-right political ideology. Building on advances in the political science literature [Bonica et al., 2014],

¹In a recent survey, 76% of Americans agreed that they were frustrated with the level of political disagreement over science news [Pew Research Center, 2022]. This frustration is bipartisan; we provide example claims of slant from both sides of the aisle. On the left, *The American Prospect* argues that the media slants their coverage of minimum wage debates: “Despite abundant empirical evidence that raising the minimum wage doesn’t lead to job loss, the idea that it does is an article of faith among right-wing economists, and all too often the media report their theological musings as fact.” On the right, many commentators argue that media outlets suppressed science related to the COVID-19 pandemic. For example, Republican Senator Ron Johnson alleged in an interview that the “mainstream media” was part of a “COVID cartel” that “suppressed and sabotaged early treatment... [and] cost hundreds of thousands of Americans their lives.”

we measure scientists’ political ideology based on their political donations.² A central challenge for previous research on politics in science has been the lack of a reliable, large-scale ideology measure.³ Our measure addresses this challenge, covering 125,284 scientists who published at least one scientific article between 2015 and 2023, and in all major scientific disciplines. We combine our measure of scientist ideology with comprehensive bibliometric information from Dimensions [Hook et al., 2018], media outlet coverage of science from Altmetric [Adie and Roe, 2013], and media outlet ideology from social media engagement patterns [Bakshy et al., 2015].

We present four sets of descriptive facts. We begin by documenting that media outlets disproportionately cover ideologically aligned scientists. A one-standard-deviation increase in ideological alignment between an outlet and a scientist is associated with a 5% increase in coverage probability. The association is primarily driven by liberal outlets being more likely to cover liberal scientists, while center and conservative outlets are both more likely to cover center scientists.

Our second result is that the quality-coverage relationship is substantially stronger than the ideological alignment-coverage relationship. We measure quality with an index of academic citations, journal impact factor, and research grants. A one-standard-deviation increase in our quality index is associated with a 34% increase in coverage probability, which is approximately five times larger than the association between ideological alignment and coverage. Strikingly, outlets are much more likely to cover high-quality papers written by *misaligned* scientists than they are to cover low-quality papers written by aligned scientists. These results suggest that news outlets’ propensity to cite ideologically aligned scientists does not lead to the coverage of low-quality science.

²The political science literature has used campaign donation data to study the ideology of actors like judges [Bonica and Sen, 2021], physicians [Bonica et al., 2014, 2020], and corporate executives [Bonica, 2016].

³To our knowledge, existing work is limited to a small subset of scientists in specific fields of science; for example Jelveh et al. [2024].

In our third set of results, we uncover a great deal of heterogeneity across scientific fields in the association between ideological alignment and coverage. The association is stronger in fields where research often engages with contemporary political debates, like the social sciences, and weaker in fields that are less obviously connected to politics, like the physical sciences. In the former fields, we estimate associations that are two to five times as large as the average estimate. In the latter fields, our estimates are small or indistinguishable from zero.

Finally, we examine how the association between ideological alignment and coverage has changed over the past decade. We observe a secular decline in the association between coverage and ideological alignment from 2015 to 2023. The only exception is a 2020 trend break, at the onset of COVID-19, when the estimated association sharply increased. To isolate the role of the pandemic, we conduct a difference-in-differences test. We compare trends in coverage of medical scientists who wrote papers on COVID-related topics to the coverage of a control group of medical scientists who never wrote papers about COVID. We find that the association between ideological alignment and coverage only increased for authors who wrote papers about COVID; for the control scientists, it continued to decline. This pattern suggests that media’s coverage of COVID-related science was more ideologically polarized, which is consistent with evidence on coverage of other COVID-related content [Hart et al. \[2020\]](#).

These last two results add nuance to the interpretation of our first two findings. Our small estimate of the association between ideology and media coverage in the aggregate is driven in part by the media’s coverage of fields where research topics are not politically charged. That media coverage across so many important scientific fields appears unaffected by ideological distortions is an encouraging finding. Our heterogeneity results, though, highlight that the magnitude of the association between coverage and ideological alignment is context-dependent. When scientific topics become relevant to political debates, ideological considerations may play a much more important role in the media coverage of research on those topics.

While our main results are descriptive, we use our heterogeneity results to explore which causal mechanisms could be driving our results. Our data are consistent with the hypothesis that outlets’ coverage of ideologically aligned *scientists* reflects a preference for the ideological content of the *science* they produce. This hypothesis explains why the association is larger in politically relevant fields of science and why it increased for authors who worked on COVID-related topics. Alternative mechanisms must explain both the heterogeneity across fields and the heterogeneity over time.

We contribute to the literature on the diffusion of scientific knowledge. While a large body of research investigates how scientific research diffuses among scientists (see [Bryan and Williams 2021](#) for a review), research on the diffusion of science to general audiences is sparser [[Yin et al., 2022](#)]. We provide a novel analysis of the role of media outlets in this process. Prior work on ideology and media coverage of science has focused on case studies of specific fields, such as climate science [[Kahan et al., 2012](#), [Chinn et al., 2020](#)]. Our estimates document the association between ideological alignment and media coverage across all major subfields of science over almost a decade.

Our results also speak to a small but emerging literature on the political economy of science and higher education. In our analysis of mechanisms, we argue that our heterogeneity results constitute indirect evidence on the role of ideology in the production of research. Previous work has documented, in specific case studies, that a researcher’s ideology correlates with the ideological content of their work. For example, studies of 238 economists [[Jelveh et al., 2024](#)], 140 legal scholars [[Chilton and Posner, 2015](#)], and 158 social scientists [[Borjas and Breznau, 2026](#)] find that research content is associated with authors’ political orientations. Our estimates, while indirect, extend the evidence base to more than 125,000 scientists across disparate fields of study.

Our large-scale measure of scientist ideology may serve as a methodological input for fu-

ture research on politics and science. It covers orders of magnitude more scientists than existing measures. It also distinguishes between center-left and far-left scientists, an important distinction given that most scientists are left-of-center [[Pew Research Center, 2009](#)]. These features should make our measure useful across a broad range of questions about the role of politics in science and higher education.⁴

Finally, we contribute to the literature on media bias. Many studies document that media outlets ideologically slant their coverage across a broad variety of domains.⁵ Reviews of this literature note that most existing studies lack a measure of the “objective” quality or importance of a news story [[Gentzkow et al., 2015](#), [Puglisi and Snyder Jr, 2015](#)], which makes it difficult to quantify the distortions that slanted coverage produces. In our context, the scientometric literature provides non-ideological quality measures, allowing us to compare the role of quality against that of ideology. We find that quality matters far more, suggesting that any distortions in coverage are small.

Sections 2–5 present the measurement strategy, descriptive empirical results, discussion of mechanisms and limitations, and conclusion, respectively.

2 Data and Measurement

2.1 Data Sources

We obtain information on researchers and scientific papers published between 2015 and 2023 from Dimensions [[Hook et al., 2018](#)]. For each researcher who published at least one article between 2015 and 2023, we collect metadata including the field of study and author institution. We measure the ideology of these researchers using the methodology discussed

⁴We have also extended our method to construct a general-purpose measure of scientist ideology covering all scientists, not only those whose research is covered by media outlets. The measure relies on publicly available data [[Bonica, 2024](#)] and can be tailored to specific applications.

⁵See, for example, [Groseclose and Milyo \[2005\]](#), [Gentzkow and Shapiro \[2010b\]](#), [Puglisi and Snyder Jr \[2011\]](#), [Chiang and Knight \[2011\]](#), [Larcinese et al. \[2011\]](#), [Martin and Yurukoglu \[2017\]](#), and [Hart et al. \[2020\]](#).

below. See Supplementary Section A.3 for further details on the sample.

We measure media coverage of scientific articles using Altmetric. Altmetric is a data informatics company that scrapes the websites of major news outlets and identifies mentions of specific scientific papers. Because Altmetric only tracks online news, our analysis focuses on the coverage of science by media outlets with at least some online presence.

We quantify the ideology of media outlets using a measure developed in [Bakshy et al. \[2015\]](#), which identifies an online media outlet’s ideology with the ideology of users who share their articles on social media platforms. The [Bakshy et al. \[2015\]](#) measure is correlated with canonical measures of outlet ideology from the literature on media bias (e.g. [Gentzkow and Shapiro \[2010a\]](#)), and covers most online outlets. Using this measure, we place the 139 outlets in our analysis sample on a scale from -1 to 1, with a negative score representing liberal ideology. See Supplementary Appendix B.1 for details on the outlets in our sample.

2.2 Measuring scientist ideology

A key challenge is measuring the ideology of scientists at scale. We utilize Campaign Finance (CF) scores, developed by [Bonica \[2014\]](#), to overcome this challenge. CF scores measure ideology based on individual donations to politicians and political campaigns with negative (positive) CF scores corresponding to liberal (conservative) donors. The CF score has been extensively validated in the political science literature [[Bonica, 2019](#), [Bonica and Sen, 2021](#), [Bonica et al., 2015](#), [Bonica and Sen, 2017](#)].

We link authors of scientific papers in our sample using a probabilistic record linking algorithm relying on name, employer, and location [[Enamorado et al., 2019](#)]. For “marginal” pairs where the algorithm is uncertain about whether a donor-author pair is a true match, we use an LLM adjudicator to make the final determination [[Peeters et al., 2023](#), [Ather, 2026](#)].⁶ We

⁶The LLM adjudicator made the final decision to include 8,424 linked scientists (7% of the analysis sample).

successfully link 125,284 authors. To validate our procedure, we manually audited the linking outcomes for 1,000 donor-author pairs, searching the web to identify true records. According to this audit, the false discovery rate of our algorithm is 5%, which implies that the vast majority of scientists in our sample were correctly matched to their CF score. In the Supplementary Section A.5, we provide details on our linking procedure and validation exercises. In Supplementary Section B, we provide key descriptive statistics.

Our measure has two key attractive features. First, to our knowledge, the coverage of our measure is an order of magnitude larger than that of existing measures of scientist ideology [Jelveh et al., 2024, Chilton and Posner, 2015, Fuchs et al., 1997, Rathbun, 2012]. Existing measures would not have been well-powered to detect the effect sizes that we observe in our results, because of sample size limitations. Second, unlike many binary measures of ideology that distinguish between Democrats and Republicans, the CF score is continuous. Since the vast majority of scientists in our sample are left-of-center (Bonica [2014]; see also Supplementary Figure S2), decisions about how to cover center-left versus far-left scientists may be an important part of how outlets slant coverage. While binary left-right measures would not allow us to observe the difference between center-left and center-right, the continuous CF score allows us to study this decision.

The application of CF scores to measure scientist ideology is a key methodological contribution of this paper. Scientist CF scores could be a valuable resource for future work on the political economy of science.

2.3 Measuring Paper Quality

We construct an index of nonideological scientific paper quality using a principal component analysis of four standard scientometric measures: academic citations, research funding, patents, and journal impact. Conceptually, this index captures scientific quality from the per-

spective of key stakeholders: academics, funders, and technologists. Of course, each of these measures suffers from well-known limitations, but we believe that together they constitute a meaningful proxy of nonideological quality. See the Supplementary Section A.3.1 for details on the construction and interpretation of the quality index, as well as descriptive statistics.

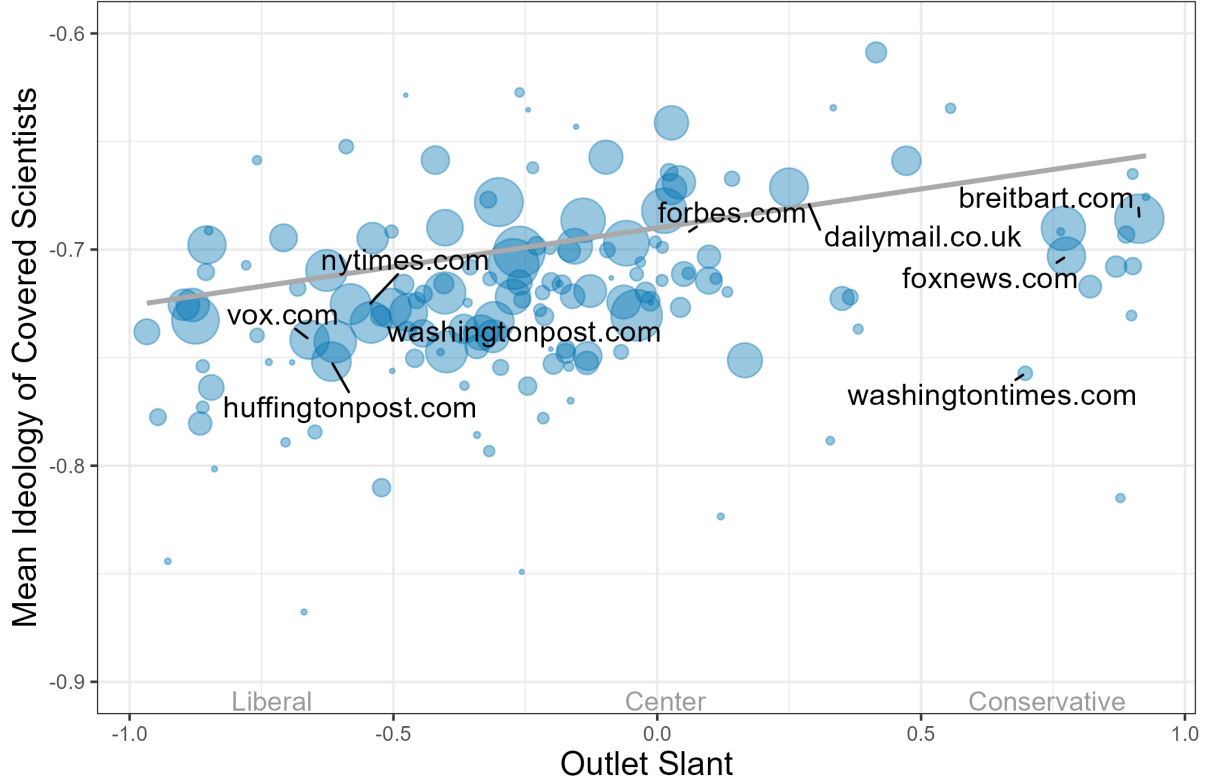
3 Facts about ideology and coverage of science

We use our novel and comprehensive data on scientists' ideology and media coverage of research papers to present four descriptive facts.

3.1 Media outlets disproportionately cover ideologically aligned scientists

Conservative (liberal) media outlets are more likely to cover research written by relatively conservative (liberal) scientists. We begin with a simple and transparent illustration. In Figure 1, we plot the mean ideology of scientists covered by a media outlet against the ideology of that media outlet. For all news outlets, the average covered paper was written by liberal scientists, because most academics are liberal. But, as news outlets become more conservative, they cover, on average, more conservative scientists.

Figure 1: Average Ideology of Covered Scientists by Outlet Slant



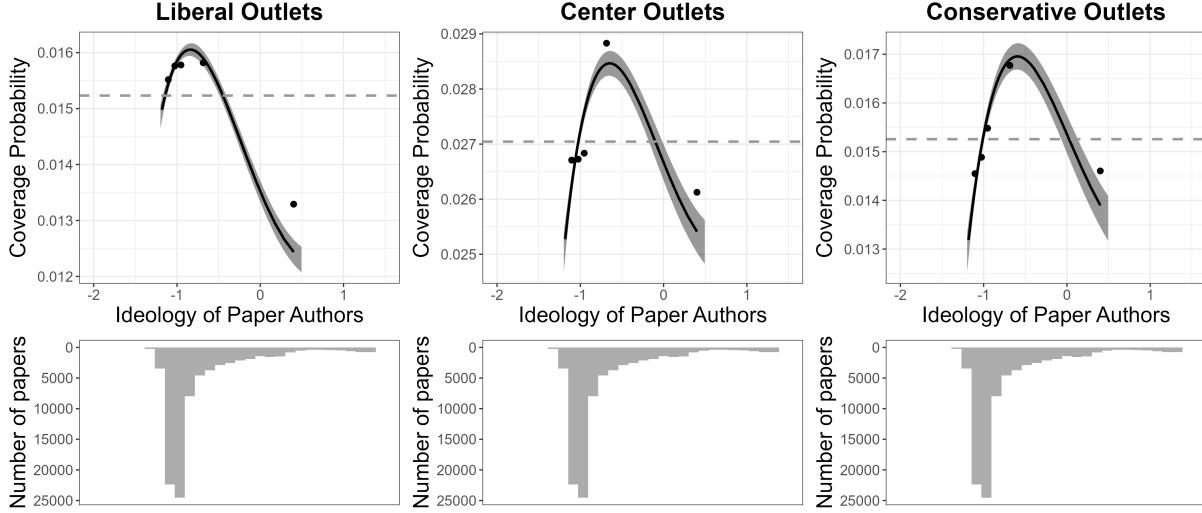
Notes: The figure presents the relationship between an outlet ideology's and the ideology of the scientists who authored papers covered by that outlet. Each observation is a media outlet. The x -axis represents outlet ideology as measured by the BMA score. The y -axis represents the mean ideology of the authors of the papers covered by the corresponding outlet. The larger points correspond to outlets with a higher number of scientific articles covered. For concreteness, some popular outlets are labeled.

Next, we estimate coverage probabilities as a flexible function of outlet ideology and scientist ideology. This approach enables us to examine nonlinearities in the coverage probability function. To do so, we use nonparametric binscatter regressions described in [Cattaneo et al. \[2024a,b, 2025\]](#). Briefly, the method estimates local polynomial regressions within different bands of paper ideology, separately for liberal, center, and conservative outlets. We linearly control for field-outlet and field-year fixed effects to adjust for an outlet's time invariant preferences for scientific fields and aggregate time trends in field coverage.

We present results in Figure 2. We find that both liberal and center outlets are most likely to cover center-left scientists. Coverage probability declines as scientist ideology deviates from the center-left in either direction. Surprisingly, even conservative outlets are more likely to cover center scientists rather than ideologically aligned conservative scientists. Outlets of different ideology mainly differ in their relative propensity to cover far-left versus center-left scientists. For liberal outlets, coverage probability falls by about 0.01 p.p. as we move to the left of the center, but for conservative outlets, it falls by about 0.04 p.p.. Since most scientists are left-of-center, this effect dominates the aggregate patterns presented in Figure 1.

In addition to providing a novel substantive insight about conservative outlets' coverage decisions, these results also highlight the importance of our continuous measure of scientist ideology. The continuous CF score is what allows us to estimate the nonlinear relationship between coverage probability and ideology. If we had instead used a binary ideology measure that only discriminated between liberal and conservative scientists, we may have drawn false conclusions about how conservative vs. liberal outlets slant their coverage.

Figure 2: Non-parametric estimates of alignment association



Notes: The figure presents coverage probability as a function of paper ideology (CF score) for liberal, conservative, and center outlets. An outlet is liberal if its BMA score is less than -0.2 , conservative if its BMA score exceeds 0.2 , and is center, otherwise. The relationship between coverage probability and ideology is estimated using a non-parametric regression, as described in Cattaneo et al. [2025, 2024a], controlling for field-year and outlet fixed effects. Observations are paper-outlet pairs. We use 10 bins, with boundaries at the deciles of the paper CF score distribution. We plot the paper CF score distribution in the bottom row. Points represent mean values for observations within each bin. Error bars represent 95% uniform confidence intervals for the conditional averages.

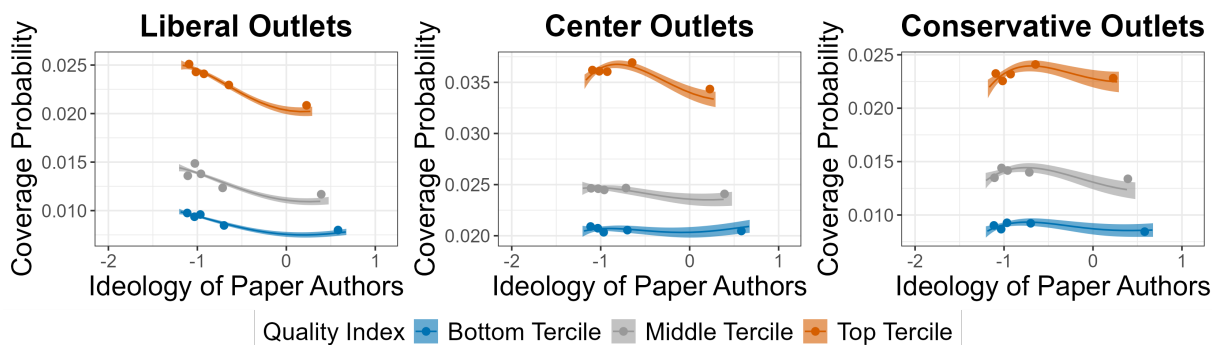
The raw correlations in Figures 1 and 2 show that media outlets preferentially cover ideologically aligned scientists. Preferential coverage of ideologically aligned research could distort the diffusion of scientific knowledge by leading outlets to cover lower-quality research. Our next analysis compares the importance of ideological alignment to the importance of objective measures of paper quality.

3.2 Ideology matters less for media coverage than (nonideological) quality

Media outlets are far more likely to cover high-quality papers than low-quality papers, regardless of the ideology of those papers' authors. In Figure 3, we present coverage probability as flexible function of the ideology of the authors similar to Figure 2, but now split the sample

of papers into terciles based on our quality index.⁷ At all levels of the quality index, the slopes of the coverage probability functions demonstrate that ideological alignment is associated with coverage. Liberal outlets disproportionately cite liberal scientists, while center and conservative outlets disproportionately avoid them. Ideology is much less important for coverage than the quality index. The quality effect is so strong that outlets are always more likely to cover high-quality, ideologically misaligned scientists than low-quality, ideologically aligned scientists.

Figure 3: Non-parametric estimates of alignment association by paper quality.



Notes: The figure presents coverage probability as a function of paper ideology (CF score) for liberal, conservative, and center outlets. Color indicates scientific quality tercile. The third tercile consists of papers with the highest quality. See Supplementary Section A.3.1 for details on the construction and interpretation of our quality measure. An outlet is liberal if its BMA score is less than -0.2 , conservative if its BMA score exceeds 0.2 , and is center otherwise. The relationship between coverage probability and ideology is estimated using a non-parametric regression, as described in Cattaneo et al. [2025, 2024a], controlling for field-year and field-outlet fixed effects. Observations are paper-outlet pairs. We use 5 bins, with boundaries at the quintiles of the paper CF score distribution. Points represent mean values for observations within each bin. Error bars represent 95% uniform confidence intervals for the conditional averages.

Compared to the broader literature on media bias, we find this result reassuring. A number of empirical papers have documented that ideology impacts the media’s coverage decisions in various domains. These analyses, though, generally do not quantitatively compare the role of ideology in coverage decisions against the role of nonideological social value or importance.

⁷One concern is reverse causality – it could be that media coverage causally increases the quality index. We show in Supplementary Table S5 that our results are qualitatively similar even when we condition on components of the quality index that are unlikely to be subject to this concern, such as research funding (which is determined before media coverage of research) and patent citations (which are determined by the technical applicability of the innovation).

One reason for this limitation is that in most settings, external proxies of social value do not exist – for example, it is challenging to quantify the “importance” of a political scandal. A unique feature of our setting is that external measures of nonideological quality have been established by the scientometric literature [Hirsch, 2005, Lehmann et al., 2006, Van Raan, 2006]. In our setting, outlets do not reduce their quality standards to cite ideologically aligned papers.

The results presented so far aggregate coverage decisions for scientific papers in all fields of science over a ten-year period from 2015 to 2023. These aggregate results may mask substantial heterogeneity, since there are differences in political relevance across scientific fields and the U.S. media industry was transformed by the shift to online and social media during our study period. We now study how our results vary across fields of science and over time.

3.3 The importance of scientist ideology differs across fields

To study heterogeneity, we estimate linear relationships between coverage and ideological alignment by different fields of science with the following regression.⁸

$$\text{Cover}_{op} = \sum_{k=1}^K \beta^k \cdot C_p^k \cdot a_{op} + \delta_{fo} + \delta_{fp(t)} + \epsilon_{op} \quad (1)$$

where Cover_{op} is an indicator of whether outlet o covered paper p . Coverage depends on *ideological alignment* a_{op} . We define ideological alignment a_{op} as the negative absolute value of the difference between the ideology score for paper p and the ideology score for outlet o . We then normalize alignment to have a mean of 0 and a standard deviation of 1.⁹ We define indicators C_p^k equal to one if paper p ’s is in field of science k . The coefficient β^k captures the association between alignment and coverage for group k . Specifically, a 1 SD increase in alignment is associated with an increase in coverage probability of β^k percentage points.

⁸Unfortunately, we do not have sufficient data to estimate non-parametric associations by field.

⁹We show in Supplementary Section D.1 that our results are qualitatively robust to alternative definitions of alignment.

We include outlet-field δ_{fo} and field-year of publication fixed effects $\delta_{ft(p)}$ as controls. We define fields according to the Australian and New Zealand Standard Research Classification (ANZSRC) Field of Research system.¹⁰ The ANZSRC categories reflect broad subject areas, such as “Biological sciences,” “Chemical Sciences,” and “Economics.”

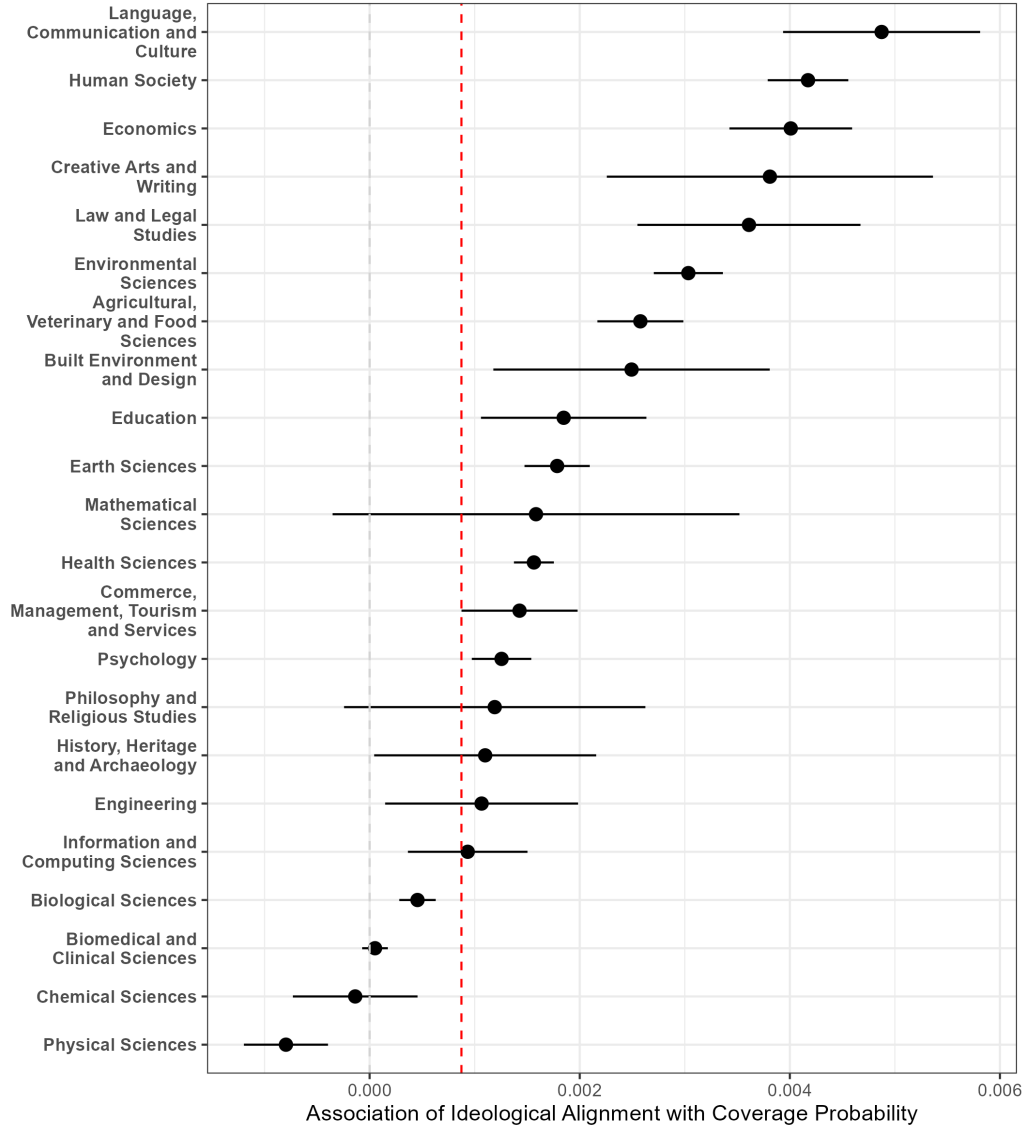
Figure 4 presents the association between ideological alignment and coverage by field of science, while also presenting, as a benchmark, the aggregate association across all fields of science. Across all fields, a 1 SD increase in ideological alignment is associated with a 0.009 p.p. increase in coverage probability.¹¹ However, there is substantial heterogeneity in the alignment association.¹² In many fields, the field-specific estimate is more than 5 times higher than the aggregate estimate. The five subject areas with the strongest alignment associations are *Language, Communication and Culture*, *Human Society* (which includes non-economics social sciences), *Economics*, *Creative Arts and Writing*, and *Law and Legal Studies*. Broadly speaking, these fields tend to engage with controversial topics and inform contemporary political debates. In contrast, for some fields alignment is not positively associated with coverage— *Physical Sciences*, *Chemical Sciences*, and *Biomedical and Clinical Sciences*. Research in these subject areas is often highly technical and may avoid engagement with controversy. Overall, these patterns suggest that ideological alignment is more strongly correlated with coverage in fields that touch on political issues. We further discuss the implications of this pattern for potential mechanisms in Section 4.1.

¹⁰Dimensions provides information on ANZSRC classification.

¹¹The mean coverage probability for an outlet paper pair is 1.9%, so this association implies a 4.7% increase in coverage probability for the average observation (Supplementary Table S5)

¹²A standard Wald test rejects the null hypothesis $\beta^1 = \dots = \beta^k$ with p -value < 0.0001 .

Figure 4: Heterogeneity in alignment associations by field of science



Notes: The figure presents the relationship between alignment and quality with media coverage for different fields of science. The y -axis represents a field of science. The x -axis presents the association between coverage propensity and alignment, controlling for outlet-field fixed effects. The association is estimated using (separate) linear probability models. The alignment measure is normalized to have a mean of 0 and a standard deviation of 1. The vertical red line denotes the average association across all fields of science.

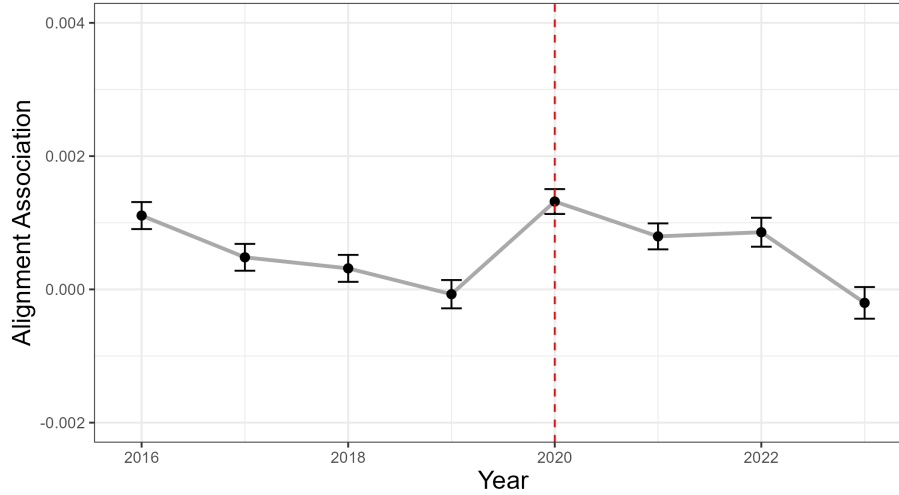
3.4 The importance of ideology declined over time, except during the COVID-19 pandemic

The media and political environments in the United States changed dramatically during our analysis period [[Cagé et al., 2020](#), [Guriev and Papaioannou, 2022](#)]. We next estimate how the association between ideological alignment and coverage changed over these years. To do so, we estimate the equation (1) with year-specific coefficients β^t , where t is the year of publication. Figure 5a presents these estimates.

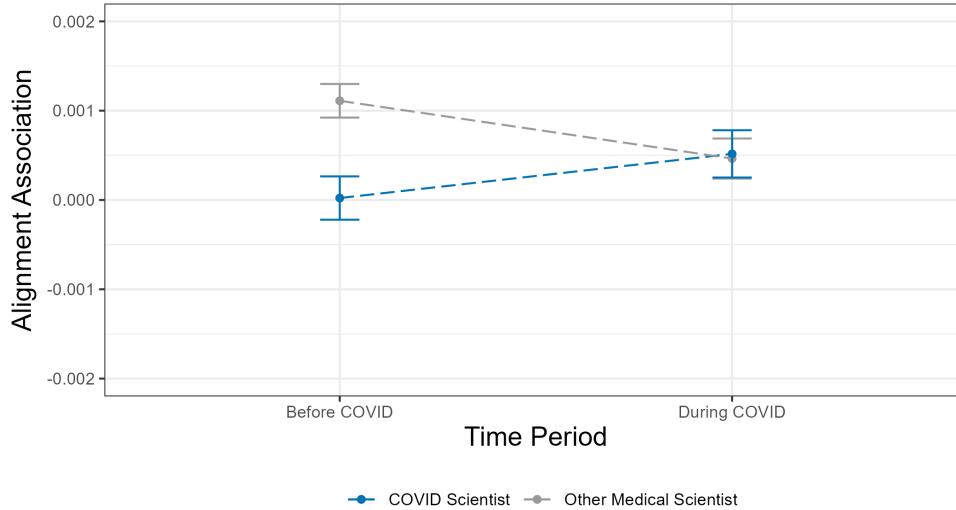
We find a secular decline in the association between ideology and coverage from 2015 to the onset of the COVID-19 pandemic. In the context of our previous discussion: the importance of ideology was small (relative to quality) to begin with, and also declining over time.

However, the pandemic was a clear trend break, with a sharp increase in the association between coverage and ideology. The COVID-19 pandemic represented an unprecedented moment in the public role of science. It generated numerous urgent, consequential questions about health and economic policy. Political leaders across the ideological spectrum frequently invoked scientific evidence to justify their positions on these contested issues. These debates were deeply polarized along partisan lines, with liberals and conservatives reaching starkly different conclusions about appropriate policy responses, and media outlets covered these issues extensively [[Hart et al., 2020](#)].

(a) The alignment association over time



(b) The impact of COVID-19 on the alignment association for COVID-19 researchers



Notes: The top figure presents changes in alignment associations for different years. The x -axis represents the time period and the y -axis represents the alignment association estimates. The bars denote 95% confidence intervals for the point estimates of β . The bottom figure presents changes in alignment associations for “COVID scientists” and “Other Medical Scientists”. A COVID Scientist is a scientist who authored at least one paper related to COVID after the onset of the pandemic. “Other Medical Scientists” are medical/health scientists who did not author any COVID-related papers. COVID-related papers are papers that appear in the CORD database (Wang *et al.* Wang *et al.* [2020]). We estimate alignment associations for these groups of scientists before and after the onset of the pandemic in 2020. Colors denote groups of scientists. The x -axis represents the time period and the y -axis represents the alignment association estimates. The bars denote 95% confidence intervals for the point estimates of β .

In our next exercise, we focus specifically on how media coverage changed for COVID-related research. We compare the role of ideology in the coverage of scientists who worked on

COVID-related research (“COVID Scientists”) relative to those medical scientists who did not (“Other Medical Scientists”), before and after the onset of the pandemic, controlling for field-outlet fixed effects.¹³ In this “difference-in-differences” comparison, we assume that absent the pandemic, coverage of scientists who ended up working on COVID would have evolved in the same way as coverage of their peers.

Figure 5b shows that after the beginning of the pandemic, ideology became differentially more important to media coverage for COVID Scientists compared to Other Medical Scientists. Before the pandemic, the alignment association was actually stronger for Other Medical Scientists. For this group, the alignment association decreased from the pre-2020 to the post-2020 period, consistent with the secular trend from Figure 5a. For COVID Scientists, though, ideological alignment became more strongly associated with media coverage. Quantifying these patterns using a standard difference in difference regression specification, we find that the change in the alignment association is significantly different across groups.¹⁴

Combined, our heterogeneity results illustrate that the importance of ideology varies across domain and time. Even if the media does not disproportionately cover ideologically aligned scientists most of the time, that can change once the scientists’ research becomes relevant to polarized public debates. These results also compel us to revisit the interpretation of our earlier finding that, on average, quality is much more important than scientist ideology for media coverage of science (Figure 3). This fact does not preclude the possibility that ideology could be much more important in coverage of areas where science touches on politically charged topics. That our aggregate estimates are so small only implies that politically salient areas of science are the exception, not the rule.

¹³We use the CORD-19 database to identify COVID-related research papers [Wang et al., 2020].

¹⁴The specification is

$$\text{Cover}_{op} = \beta^0 \cdot a_{op} + \beta^1 \cdot \text{Post}_p \cdot a_{op} + \beta^2 \cdot \text{COVIDScientist}_p \cdot a_{op} + \beta^3 \cdot \text{Post}_p \cdot \text{COVIDScientist}_p \cdot a_{op} + \delta_o + \delta_p + \epsilon_{op} \quad (2)$$

The estimate of β^3 is 0.0017 (0.0002).

4 Potential Mechanisms and Limitations

4.1 What drives the alignment association?

Next, we discuss plausible mechanisms that could generate the patterns of association between ideology and coverage. We reiterate that the results in Section 3 are descriptive patterns, not causal relationships between ideological alignment and media coverage; therefore, this discussion is exploratory.

We separate potential mechanisms into two classes. First, outlets may prefer to cover articles with content that align with their viewpoints, and that ideologically aligned scientists produce ideologically aligned content. This *ideological content mechanism* would be consistent with a large economics of the media literature documenting that outlets slant the content of their coverage [Puglisi and Snyder Jr, 2015]. Alternatively, conservative and liberal outlets may differ in their propensity to cover scientists with different ideologies, holding the content of research fixed. Such a “direct effect” of scientist ideology could arise for several reasons. For example, scientists and journalists who share political views may be more likely to form professional connections.¹⁵

The heterogeneity across fields supports the content mechanism. In Figure 4, we found that ideological alignment matters more for coverage in fields of science that regularly engage with contemporary political debates and include ideologically relevant content. By contrast, ideological alignment mattered less for fields of science with relatively little political relevance—how can a paper about the latest experiment at the Large Hadron Collider be “conservative” or “liberal?”. The ideological content mechanism is a parsimonious explanation for this pattern.¹⁶

¹⁵There is evidence that people prefer to form social connections with ideologically aligned peers, see e.g. McPherson et al. [2001].

¹⁶For an alternative mechanism to explain this pattern, an omitted factor must affect coverage decisions more in politically relevant than politically irrelevant fields. For example, stronger ideological homophily in social versus physical sciences would explain the pattern.

In a similar vein, our results on the impact of COVID pandemic also support the ideological content mechanism. Once the pandemic started, the research written by COVID Scientists, on topics like masking and virus transmission, became more politically relevant. This increase in the political relevance of content plausibly resulted in increased association between coverage and ideological alignment. Alternative explanations would require media outlets to differentially consider scientist ideology exactly at the onset of the COVID-19 pandemic.

In summary, our results on heterogeneity by field of science and on the impact of the COVID pandemic are both consistent with the ideological content mechanism. These complementary sources of evidence are based on different assumptions. Alternative explanations for the field heterogeneity results would require omitted factors to vary between science fields. Alternative explanations for the COVID results would require omitted factors to change differently for COVID Scientists at the onset of the pandemic. Although this analysis is by no means conclusive, we view our results as suggestive evidence for the importance of the ideological content mechanism.

To the extent that the ideological content mechanism is important, our results also provide novel indirect evidence about the role of ideology in the production of research. Media outlets disproportionately cover research written by scientists who made personal donations to politically aligned candidates. If these coverage decisions are driven by ideological content, then scientists who hold left-leaning political views must produce research with more left-leaning content. A relatively small literature has studied how scientists' personal political views affect research output. Most studies have focused narrowly on fields of science where ideological content can be directly measured, such as economics [[Jelveh et al., 2024](#), [Borjas and Breznau, 2026](#)] and law [[Chilton and Posner, 2015](#)]. Our approach, by contrast, provides evidence on ideology and the production of research across all fields of science.

4.2 Limitations

Our results should be interpreted in light of several important caveats and limitations. First, our analysis is limited only to scientists who make political contributions in U.S. campaigns. These scientists are not a representative sample of academic researchers—they are probably more politically engaged than the average researcher—and our results may not generalize beyond this analysis sample. Second, research quality is high-dimensional; our measures of quality necessarily only capture some important dimensions. Third, our analysis focuses narrowly on the “filtering” decisions outlets make about *whether* to cover papers. Such filtering decisions are tractable for empirical analysis and are an important dimension by which the media slants its coverage [[Gentzkow et al., 2015](#)]. We emphasize, though, that we do not present any results about *how* media outlets cover work by different scientists.

In addition, our evidence on mechanisms, including our arguments that media outlets may make coverage decisions based on ideological content and that the ideological content of research may be impacted by the authors’ political views, is suggestive and indirect. We hope that future research directly tests these important hypotheses, potentially facilitated by our novel large-scale measure of scientist ideology.

5 Conclusions

This paper developed a novel measure of scientist ideology and used it to study the role of ideology in media coverage of research. We found that media outlets disproportionately cover research written by ideologically aligned scientists, but objective measures of research quality matter much more for coverage decisions than scientist ideology. The association between ideological alignment and coverage is larger in fields of science that engage with contemporary political debates, and it became larger for scientists who wrote papers related to the COVID-19

pandemic. These results are consistent with a mechanism where scientists create research with ideological content, and news outlets make coverage decisions that depend on this ideological content.

While we have interpreted our results on the relative importance of paper quality versus ideological alignment optimistically, we close with a note of caution. Our results on heterogeneity across fields and over time illustrate that the media's preference to cover ideologically aligned scientists is not constant. Instead, it depends on the domain of research, incentives for scientists, and on the political climate. If more scientific topics become politically charged, ideological distortions in media coverage of science could become more powerful.

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